# LESSON 8.2
# SOLVING REAL-WORLD, MULTI-STEP EQUATIONS

## LESSON INTRODUCTION

**MA.8.AR.2.1** Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

### LEARNING TARGETS

- I can solve multi-step linear equations from a real-world context.

### BENCHMARK COHERENCE

**BUILDING ON**

**MA.7.AR.2.2** Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

**WORKING TOWARD**

**MA.912.AR.1.2** Rearrange equations or formulas to isolate a quantity of interest.

**MA.912.AR.2.1** Given a real-world context, write and solve one-variable, multi-step linear equations.

### ESSENTIAL QUESTIONS

- How can algebraic equations be used to solve a real-world problem?
- How do you solve an equation in one variable?

### LESSON NARRATIVE

Students will solve multi-step equations representing real-world contexts, including interpreting the solutions in terms of the context. The lesson continues to reinforce the fact that there are different pathways to solving equations. Students will discuss different approaches with their partners and begin to consider how the structure of an equation might inform their approach as they continue to build flexible thinking.

### COMPONENT SUMMARY

**8.2.1 WARM-UP** *(~5-10 MIN.)*

Create a numerical sequence

**8.2.2 COLLABORATION** *(10 MIN.)*

Solve an equation in multiple ways

**8.2.3 GUIDED INSTRUCTION** *(10-15 MIN.)*

Solve problems using multi-step equations

**8.2.4 YOUR TURN!** *(10 MIN.)*

Solve multi-step equations with a real-world context

**8.2.5 WRAP-UP** *(~5 MIN.)*

Solve a real-world equation

# RESOURCES

### GLOSSARY

Key Terms: N/A

Additional Terms: N/A

### MATERIALS

- (Optional) Highlighters or colored pencils

### ADDITIONAL PREPARATION

- N/A

# TEACHER TIPS

### COMMON MISCONCEPTIONS

- Students may forget to distribute to all of the terms in parentheses when applying the distributive property.
- Students may struggle to carry a negative with a number when performing an operation.

### THINGS TO CONSIDER

- Consider using information from these first two lessons to identify students who may need additional support as the difficulty of the equations they are solving increases in the subsequent lessons.

### LITERACY AND LANGUAGE ROUTINES

**Compare and Connect**

To foster students' meta-awareness of different mathematical approaches and representations, encourage students to notice and compare the methods that different students use when solving problems. Emphasize that problems can be solved in multiple ways.

### MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

**MA.K12.MTR.4.1 Discussion and Reflection**

Encourage students to think about multiple ways to solve and represent each problem. Some students may benefit from using one approach instead of another. Encourage students to explain why they are using a specific approach to solve a problem and why they think it is the best method to use in a specific situation.

# DIFFERENTIATE INSTRUCTION

Consider providing a table to record each step of work so as to keep steps neat and avoid spilling over into another step. Some problems could be scaffolded by having the notes on a study guide with the problems in the lesson partially written, where students could fill in the missing parts for each step. Some students may benefit from using highlighters or colored pencils to color code like terms and or the similarities in the warm up sequences.

# 8.2.1 WARM-UP: HAILSTONE SEQUENCE

## TEACHER MOVES

**Compare and Connect**

After students have had a few minutes to create their own sequence, the teacher could consider writing example sequences created by students on the board that could be used as additional evidence when students make their conjectures. If doing this, teacher should wait until most students have completed question 1 before writing any examples on the board.

Encourage students to note similarities and differences between the sequences and their starting values.

### 8.2.1 Warm-Up: Hailstone Sequence

Hailstones move through the atmosphere before falling to the earth. They start in a cloud as drops of rainwater, then push higher into the atmosphere where they freeze before eventually falling back to Earth.

The up-and-down pattern led mathematicians to call some sequences of numbers a Hailstone sequence. A Hailstone sequence follows two rules.

- If a number is even, divide it by $2$.
- If a number is odd, multiply it by $3$ and add $1$.

The sequence shown is one example of a Hailstone sequence.

$$20, 10, 5, 16, 8, 4, 2, 1, \ldots .$$

1. Create your own Hailstone sequence by choosing a starting value, and then applying the Hailstone rules to each new term.
   - If the previous term is even, divide it by $2$.
   - If the previous term is odd, multiply it by $3$ and add $1$.

   *Answers will vary.* $32, 16, 8, 4, 2, 1, 4, 2, 1$

2. Compare your sequence with the example sequence and record any observations you made.
   *Answers will vary. My sequence has part of the same series of numbers* $(16, 8, 4, 2, 1)$ *as the example sequence.*

## TEACHER MOVES

Encourage students to look for similarities among the sequences to develop generalizations that can be used to make a conjecture.

## QUESTIONING

Encourage students to explore other sequences, such as the Fibonacci Sequence. Create an opportunity for students to learn about the Fibonacci Sequence, how it works, and its applications in nature.

A conjecture is an idea that you think might be true, but do not know for sure.

3. Based on your observations, make a conjecture about all Hailstone sequences.
   *Answers will vary. The sequence will always contain* $2$ *and then* $1$.

In 1937 Lothar Collatz, a mathematician, proposed that for any starting value of a Hailstone sequence, the sequence will always reach 1. Since the conjecture cannot be proven or disproven, it is now called the Collatz Conjecture and is classified as an unsolved problem in mathematics.

### 8.2.2 Collaboration: How Many Cookies?

1. A large container of cookies was donated for the school bake sale. Two cookies were taken out of the container and set aside so that the remaining cookies could be evenly divided among four plates, each with a dozen and a half cookies.

   The situation can be represented by the equation $\frac{1}{4}(c - 2) = 18$, where $c$ is the number of cookies in the large container at the start.

   Three students completed the first step in solving this equation.
   - Joel said the first step in solving this equation is to distribute the $\frac{1}{4}$.
   - Elena said she was going to first add $2$ to both sides to solve the equation.
   - Nichele said she was going to start by multiplying both sides of the equation by $4$.

   a. Work with your partner to finish solving the equation using each student's suggested first step, shown in blue.

| Joel | Elena | Nichele |
|---|---|---|
| $\frac{1}{4}(c - 2) = 18$ | $\frac{1}{4}(c - 2) = 18$ | $\frac{1}{4}(c - 2) = 18$ |
| $\frac{c}{4} - \frac{2}{4} = 18$ | $\frac{1}{4}(c - 2 + 2) = 18 + 2$ | $4 \cdot \frac{1}{4}(c - 2) = 18 \cdot 4$ |
| $\frac{c}{4} - \frac{2}{4} + \frac{2}{4} = 18 + \frac{2}{4}$ | $\frac{1}{4}c = 20$ | $c - 2 = 72$ |
| $\frac{c}{4} = 18\frac{2}{4}$ | $(4)(\frac{c}{4}) = (4)(20)$ | $c - 2 + 2 = 72 + 2$ |
| $(4)(\frac{c}{4}) = (4)(\frac{74}{4})$ | $c = 80$ | $c = 74$ |
| $c = 74$ | | |

   b. Identify the student who made an error in their first step and explain what error they made.
   *Elena made an error when she added $2$ to both sides in the first step. Adding negative $2$ in parentheses when a factor is to be distributed is not the same as adding $2$.*

   c. Complete the statement to interpret the solution in this context.

   There were <u>74</u> cookies in the large container at the start of the bake sale.

# 8.2.2 COLLABORATION: HOW MANY COOKIES?

## DIFFERENTIATE INSTRUCTION

Consider providing a table or scratch paper to record each step of work. This will help keep students steps neat and avoid spilling over into another step. This strategy may assist students who struggle with a string of numbers and variables, especially students with dyslexia, dysgraphia, and dyscalculia.

## ENRICHMENT

Why does $\frac{1}{4}c = \frac{c}{4}$? Why does Joel rewrite the fraction this way after distributing the 4?

## ENRICHMENT

Consider asking the following probing questions, as needed:

- What was Joel's first step?
- How did Nichele begin solving the equation?
- What are the benefits of Joel's approach?
- What do you like about Nichele's method?

## TEACHER MOVES

Encourage students to analyze and discuss the different ways that Joel and Nichele approached the problem. Encourage students to generalize when they think dividing first may work well and in what situations they think distributing first would work best.

 

# 8.2.3 GUIDED INSTRUCTION: SOLVING MULTI-STEP EQUATIONS IN REAL-WORLD CONTEXTS

## TEACHER MOVES

Guide students to understand that looking at the coefficients and constants in an equation can help determine the best approach to use when solving an equation. Understanding equivalency of expressions and equations empowers students to have flexible thinking. Incorporating different equivalent expressions and equations helps model this flexible thinking for students.

## ENRICHMENT

- What is a benefit of using Paul's method?
- Why do you think he chose to begin by multiplying by 100?
- What are the drawbacks of Paul's approach?

### 8.2.3 Guided Instruction: Solving Multi-Step Equations in Real-World Contexts

1. Uncle Frank loves riddles. Uncle Frank tells his nephews, "I have 3 times as many dimes as quarters. I have $20$ more nickels than quarters. I have $\$6.40$ total. How many of each type of coin do I have? Whoever can solve my riddle will get my coins."

   The riddle can be represented by the equation $0.25m + 0.10(3m) + 0.05(20 + m) = 6.40$, where $m$ represents the number of quarters Uncle Frank has.

   Paul and Matthew both agree they should apply the distributive property first. The resulting equation is shown.

   $$0.25m + 0.30m + 1.00 + 0.05m = 6.40$$

   Paul said the next step is to multiply both sides by $100$.

   a. Solve the equation using Paul's method.

   | Paul's Method |
   |---|
   | $0.25m + 0.30m + 1.00 + 0.05m = 6.40$ |
   | $100(0.25m + 0.30m + 1.00 + 0.05m) = 100(6.40)$ |
   | $25m + 30m + 100 + 5m = 640$ |
   | $60m + 100 = 640$ |
   | $60m + 100 - 100 = 640 - 100$ |
   | $\frac{60m}{60} = \frac{540}{60}$ |
   | $m = 9$ |

   b. What do you notice about the resulting values after multiplying by $100$?
   *The decimal numbers are gone.*

Matthew said it would have been better to first simplify the left side of the equation by combining like terms, then solve.

c. Solve the equation using Matthew's method.

| Matthew's Method |
|---|
| $0.25m + 0.30m + 1.00 + 0.05m = 6.40$ |
| $0.6m + 1.00 = 6.40$ |
| $0.6m + 1.00 - 1.00 = 6.40 - 1.00$ |
| $\frac{0.6m}{0.6} = \frac{5.40}{0.6}$ |
| $m = 9$ |

d. Matthew believes that Paul's method is incorrect. Discuss with your partner if Paul's method was correct or incorrect. Summarize your discussion.
*Paul's method is also correct. It works as long as you multiply both sides by the same number,* $100$.

e. Explain whose method you prefer for solving this equation.
*Answers will vary. Paul's method gets rid of the decimal numbers sooner, but Matthew's method has fewer steps.*

f. Use the solution to the equation to solve Uncle Frank's riddle and determine the number of quarters, dimes, and nickels he has.

| **Quarters,** $m$ | **Dimes,** $3m$ | **Nickels,** $20 + m$ |
|---|---|---|
| $m = 9$<br>*He has* $9$ *quarters.* | $(3)(9) = 27$<br>*He has* $27$ *dimes.* | $20 + 9 = 29$<br>*He has* $29$ *nickels.* |

# 8.2.3 GUIDED INSTRUCTION: SOLVING MULTI-STEP EQUATIONS IN REAL-WORLD CONTEXTS (CONTINUED)

## ENRICHMENT

- What is a benefit of using Matthew's method?
- Why do you think he chose to begin by combining like terms?
- What are the drawbacks of Matthew's approach?

## TEACHER MOVES

Reinforce the importance of assessing the feasibility of answers. Encourage students to substitute the solution back into a problem to determine if it is correct.

## ENRICHMENT

- How does the number of quarters relate to the solution of the equation?
- How would you determine the number of dimes? How would you figure out the number of nickels?
- How does the expression for each coin relate to the wording of the problem?

# 8.2.4 YOUR TURN!

**ENRICHMENT**

Monitor whether students are relating the solutions to the contexts. Try asking students: what does the solution mean in context?

**8.2.4 Your Turn!**

1. Carla and some friends went to a movie. They spent $\$25.95$ on snacks and each ticket cost $\$9.50$. In total, the friends spent $\$54.45$ for the night out. The equation $54.45 = 9.50t + 25.95$ can be used to model the situation, where $t$ represents the number of tickets purchased.

   Solve the equation to determine how many tickets were purchased.

   $54.45 = 9.50t + 25.95$

   $54.45 - 25.95 = 9.50t + 25.95 - 25.95$

   $\frac{28.5}{9.50} = \frac{9.50t}{9.50}$

   $t = 3$

   *They purchased 3 tickets.*

2. Jackson works as a manager in an office. He is scheduled to work $40$ hours each week, and for every hour worked over $40$, he earns one-and-a-half times his normal hourly pay. If Jackson worked $45$ hours one week and earned $\$1{,}306.25$, use the equation $p(40 + 7.5) = 1{,}306.25$ to find $p$, the hourly pay that Jackson earns.

   $p(40 + 7.5) = 1{,}306.25$

   $47.5p = 1306.25$

   $p = \$27.50$

   *Jackson earns $27.50 per hour.*

### 8.2.5 Wrap-Up: Ticket Out the Door

1. Kiran and his family are planning to attend the Central Florida Fair. They have a coupon for $\$1.50$ off each ticket. They paid $\$46.50$ for all $6$ tickets. Find the normal cost of one ticket without using the coupon, $c$, using the equation, $6(c - 1.50) = 46.50$.

   $6(c - 1.50) = 46.50$

   $c - 1.50 = 7.75$

   $c = 9.25$

   *The normal cost of one ticket is* $\$9.25$.

# 8.2.5 WRAP-UP: TICKET OUT THE DOOR

**DIFFERENTIATE INSTRUCTION**

Observe students as they work through 8.2.5 Wrap-Up. Note steps in which students may struggle and specific issues they encounter. Use these observations to plan for differentiation in future instruction as needed.